

**ASSESSMENT OF CARDIOVASCULAR RISK USING WEARABLE
DEVICE DATA: A PYTHON-BASED ANALYSIS OF
PHYSIOLOGICAL HEALTH INDICATORS**

Abstract

In this study, statistical associations between the physiological and behavioural factors extracted from wearable devices and cardiovascular risks calculated based on secondary health data will be explored. Originally, there were 5,000 records for 1,000 participants, with five records for each participant (measurement level). After data quality control and summarisation at the level of participants, 1,000 records were analysed. The following analytical methods, such as descriptive statistics, normality assessment, Spearman's rank correlation, Kruskal-Wallis's test and Chi-Square Analysis, will be used to test the relationships between selected factors and cardiovascular risk. In the sample of participants, there were 374 (37.4%) participants at low risk of cardiovascular events, 603 (60.3%) at moderate risk, and only 23 (2.3%) at high risk. According to Spearman's analysis, there is a positive and strong relationship between the cardiovascular risk score and age ($\rho = .887$), average heart rate ($\rho = .842$), resting heart rate ($\rho = .833$), systolic blood pressure ($\rho = .832$) and diastolic blood pressure ($\rho = .828$) (all $p < .001$). BMI demonstrates a statistically significant positive correlation, while SpO₂ shows a weak negative correlation. According to the Kruskal-Wallis test, statistically significant differences in risk categories are seen for age, BMI, resting heart rate, average heart rate, systolic and diastolic blood pressure and daily steps. But we need to remember the relatively small size of the group with a high risk. Gender was not statistically significantly related to risk, $\chi^2 (2) = 1.667$, $p = .434$. Physiological factors demonstrate much stronger relationships with the cardiovascular risk score in the dataset compared to behavioural ones.

Code Availability

All Python scripts used for data preprocessing, statistical analysis, and visualisation in this paper are publicly available at the GitHub University repository. The repository holds the entire workflow necessary for reproducing the statistical analysis and figures reported in this paper.

Repository: <https://github.coventry.ac.uk/Yerramsetp/Cardiovascular-Risk-Wearable-Analysis.git>.

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Chapter 1: Introduction

1.1 Introduction

Cardiovascular disease is one of the major fields of healthcare research that is relevant for the early detection of risk and the application of preventive measures. The use of wearable health devices allows researchers to collect information regarding various physiological and behavioural indicators, including heart rate, blood pressure, physical activity, duration of sleep, and oxygen saturation in the blood. The collected data can help to get a better idea about cardiovascular risk that goes beyond occasional measurement by a medical professional. In this context, the current study will examine the correlations between health indicators recorded with the use of wearable devices and cardiovascular risk scores.

1.2 Background of the Study

Cardiovascular disease (CVD) still poses one of the largest health issues since cardiac diseases progress gradually without any visible symptoms. It is vital to detect people with high cardiovascular risk early to be able to apply preventive interventions and modifications in their lifestyle and conduct clinical evaluations. The traditional assessment of the risk for CVD usually uses repeated clinical measurements and interviews. Even though this method is still used, it gives episodic data about patients' physiology and behaviour patterns.

However, with the appearance of wearable health technologies, there are new possibilities for more regular and evidence-based health monitoring. By using smartwatches, fitness trackers and other wearable sensors, one can continuously monitor physiological and behavioural variables,

including resting and average heart rate, number of steps made per day, minutes of activity, hours of sleep, oxygen saturation (SpO₂), energy expenditure and physical activity (Cafolla 2024, August). Adding this data to demographic data and the indicators of cardiovascular risk such as age, BMI and systolic and diastolic blood pressure, one can obtain a full picture of potentially influential factors.

Hence, wearable data are becoming increasingly useful in terms of preventive medicine and remote health monitoring. Nevertheless, the collection of vast amounts of data does not imply that the data that are more related to the indicators of interest will automatically be identified. For that purpose, one needs to conduct statistical analysis to identify the most influential data. This study aims to fill this gap by analysing wearable and physiological data of participants and finding statistically significant associations between the variables and cardiovascular risk indicators.

1.3 Problem Statement

In assessing cardiovascular risk, there have been routine clinical examinations that give valuable data, but that may not necessarily be indicative of changes in physiological state and behaviour of individuals on an ongoing basis. Wearable devices can help bridge this gap and provide useful data like heart rate, physical activity, sleep time, oxygen saturation and energy expenditure. But how these data indicators, which can be collected on an ongoing basis through wearable devices, are related to cardiovascular risks is a question that needs systematic research. Only when it is established that the wearable data indicators have a statistical relationship with cardiovascular risks will it become possible to ascertain what wearable data indicators can say about cardiovascular risks. The current study tries to address this challenge using analysis of 5,000

measurement-level data from 1,000 individuals and the relationships between their cardiovascular risk scores and physiological, demographic and lifestyle indicators.

1.4 Research Rationale

With the growing use of wearable devices, it has become possible to monitor certain physiological and behavioural indicators on a constant basis without using a conventional clinical setting. Measures such as heart rate, blood pressure, level of physical activity, time spent sleeping, oxygen saturation and energy expenditure can give much useful information about one's cardiovascular system state. Nevertheless, a high volume of wearable data cannot guarantee the existence of certain connections that link some indicators with the risk of having any cardiovascular diseases. That is why there is a necessity to do some systematic investigation using data analysis.

This research becomes legitimate because of the potential for using wearable data in the identification and monitoring of cardiovascular risk. Statistically studying the relationship between certain health indicators and cardiovascular risk scores makes it possible to prove which of them have the closest connection. In addition, this study fills the gap in the literature that is connected with the simultaneous consideration of several health indicators.

1.5 Research Aim

The objective of the current research study is to understand the associations between wearable health parameters and cardiovascular risks through statistical analysis of physiological, behavioural, and demographic variables and determine which of these parameters are highly correlated with the cardiovascular risk scores and groups within the chosen wearable health data set.

1.6 Research Objectives

To achieve the research objective, the research will set the following specific objectives as follows:

1. To study and preprocess the data on wearable health factors that include demographic, physiological, lifestyle, sleep, energy expenditure and cardiovascular risk factors so as to make it suitable for statistical analysis.
2. To study the correlation between wearable health factors and the risk scores of cardiovascular diseases using correlation analysis techniques and to identify the significantly correlated factors.
3. To study and compare physiological and lifestyle factors among individuals in low, moderate and high-risk categories in relation to cardiovascular disease risks to see if there are significant differences between them.
4. To identify the factors that are significantly correlated with cardiovascular risk and to assess the relevance of such factors to data-based cardiovascular risk screening.
5. To identify any relationship between cardiovascular risk category and gender.

1.7 Research Question and Hypotheses

The research questions guiding the study include:

RQ: What relationships can be found between wearable health metrics and cardiovascular risk of the participants in the selected dataset?

To test the above research question, statistical hypotheses were put forward for each predictor under analysis. In general, they are as follows:

H_0 : No statistically significant monotonic relationship exists between the selected wearable health metric and cardiovascular risk score.

H_1 : There is a statistically significant monotonic relationship between the selected wearable health metric and cardiovascular risk score.

The hypotheses were tested for age, BMI, resting heart rate, average heart rate, average steps, active minutes, sleep time, SpO_2 , burned calories, resting energy expenditure, active energy expenditure, systolic blood pressure, and diastolic blood pressure. Statistical decisions were made at $\alpha = 0.05$. H_0 was rejected if $p < 0.05$.

1.8 Significance of the Study

Firstly, this work is of interest for the field of wearable-health studies because it empirically examines the connection between some regularly measured physiological and lifestyle variables and the probability of having cardiovascular disease. From an academic perspective, the contribution of this work is related to better understanding the relationship between variables such as age, heart rate, blood pressure, BMI, physical activity, sleep, and SpO_2 and cardiovascular risk score in the wearable-health dataset. The findings especially emphasise age, average heart rate, resting heart rate, and blood pressure as significant variables that can be explored further. From a practical point of view, the findings show that wearable health data have the potential to be useful for monitoring cardiovascular risk continuously and for early recognition of people who require additional examination.

1.9 Scope and Delimitations

The current study was carried out to analyse health data that can be used for studying connections between different physiologic and lifestyle factors and cardiovascular risks. The initial data included 5,000 records on measurements per 1,000 people for 18 different characteristics collected in 2025. Data analysis was performed at the person level and included 1,000 observations divided into groups depending on their cardiovascular risk score (low, medium, and high) (Opoku-Gyamfi 2025). The following characteristics were studied: age, BMI, heart rate, blood pressure, SpO₂, physical activity, sleep, calorie expenditure, and energy expenditure in connection with the cardiovascular risk score.

1.10 Dissertation Structure

The current study is presented in five chapters. The first chapter introduces the background of the research, research problem, objectives, research questions, and significance of the research. The second chapter presents the review of the literature on wearable health technologies and cardiovascular risk assessment. The third chapter introduces the research methodology and statistical techniques used. Chapter 4 presents and analyses the findings related to the descriptive, correlation, group comparison, and hypothesis testing. Finally, the fifth chapter concludes the research by analysing the findings, research question, and research objectives.

Chapter 2 - Literature Review

2.1 Introduction

The relevance of wearable devices in cardiovascular care has been enhanced due to the provision of real-time tracking of physiological and behavioural factors of health (Li et al., 2025; Moshawrab et al., 2023). From the current literature, these wearable devices can contribute to the process of early risk identification for cardiovascular problems, remote monitoring and evidence-based clinical decision-making. This chapter will critically examine the existing research about wearable-based cardiovascular monitoring, emphasising physiological and behavioural factors of health.

2.2 Wearable Devices in Cardiovascular Health Monitoring

The significance of wearables in cardiovascular healthcare has been increasing because of the ability of these devices to continuously and non-invasively monitor physiological and behavioural signs beyond traditional clinical settings. For example, smartwatches and wearable devices can measure heart rate, physical activity, energy expenditure, and other health-related metrics, thus allowing continuous tracking of the state of a cardiovascular system rather than its evaluation at moments in time. Li et al. (2025) noted that continuous physiological monitoring helps provide timely clinical interventions, personalised treatment, and decision-making about health services. Likewise, Moshawrab et al. (2023) mentioned that wearable devices able to monitor heart rate and electrocardiogram signs could help detect cardiovascular problems in the early stages and manage patients remotely. Nevertheless, the quality of wearable data and their clinical significance may depend on issues related to the reliability of measurements, interoperability, and device-related problems (Canali et al., 2022). Thus, wearables represent excellent possibilities to monitor the

cardiovascular system; however, the data should be analysed carefully before using them for risk evaluation purposes.

2.3 Physiological Indicators of Cardiovascular Risk

Physiological markers, as obtained through wearable and digital health technologies, are significant for assessing cardiovascular health. Age is a basic risk-related feature since cardiovascular functions change with age, while other metrics, such as heart rate and blood pressure, offer direct insights into cardiovascular functioning. Continuous monitoring of physiological data, provided by wearable technologies, can facilitate early detection of cardiovascular abnormalities and preventive intervention (Li et al., 2025). Furthermore, Li et al. state that timely cardiovascular information can improve personalised care and clinical decision-making.

Resting and average heart rate should be considered since heart rate is measured by wearable devices regularly. As mentioned by Moshawrab et al. (2023), evidence shows that smart wearables typically assess heart rate along with other physiological parameters and can detect cardiovascular abnormalities. Blood pressure is equally important for assessing cardiovascular problems; hence, systolic and diastolic pressure are other indicators that can be considered along with heart rate when assessing cardiovascular risk.

BMI is another measurement, which reflects body composition and can be used together with cardiovascular measures to obtain a more comprehensive overview of individual risk. SpO₂ offers information about the blood oxygen level and extends the list of physiological markers that can be obtained through health-monitoring technologies. Still, measurements from wearable devices must be carefully interpreted because the quality of data, interoperability, and reliability of devices may

affect their clinical usefulness (Canali et al., 2022). Although sophisticated analytical models can reveal complicated associations between physiological indicators, low interpretability may limit their clinical significance (Huang et al., 2022). Therefore, age, heart rate, blood pressure, BMI, and SpO₂ should be examined through transparent statistics to determine which markers are significantly associated with cardiovascular risk.

2.4 Behavioural and Activity Indicators

Indicators related to physical activity such as the number of steps per day, number of active minutes per day, number of hours spent sleeping and energy expenditure are useful for assessing lifestyle factors related to cardiovascular disease. Wearable technologies allow us to have continuous monitoring of physical activity and physiological behaviour, which can be beneficial in preventive care and telemedicine (Moshawrab et al., 2023). In a similar way, Sultana et al. (2025) argued that continuous monitoring through wearables can support preventive healthcare due to continuous observation and early detection of changes in health status.

Nevertheless, the value of behavioural measurements is determined by their reliability. Fuller et al. (2020) tested the reliability of the available commercial wearable devices in terms of steps, energy expenditure and heart rate measurements. As a result, they concluded that measurement reliability depends on the conditions of the process. Moreover, Canali et al. (2022) considered data reliability and quality as important issues that can influence the clinical value of wearable information. Thus, it is possible to assess steps, activity, sleep and energy expenditure together with physiological measurements, considering the quality of measurement through wearables.

2.5 Critical Review of Recent Empirical Studies

In recent years, more empirical research has been done with the use of physiological and behavioural data collected through wearable technology in cardiovascular and cardiometabolic assessment. For instance, Browne et al. (2021) performed a 12-month randomised controlled trial using the biometric ring to demonstrate that wearable-technology-enabled feedback could enhance sleep and exercise behaviours. The study included measures such as step count, heart-rate variability and sleep parameters, thus showing the potential of wearables to assess multiple health aspects. Nevertheless, focusing on behavioural change instead of direct risk assessment does not allow determining which physiological parameters are significantly related to cardiovascular risk (Browne et al., 2021).

The use of wearable technology in haemodynamic indicator evaluation is demonstrated by more recent empirical findings. For example, a study published in 2023 in *Nature Biomedical Engineering* used wearable electrocardiography and multispectral photoplethysmography to monitor cardiovascular measures associated with the regulation of blood pressure in healthy people, as well as in those suffering from hypertension and those who recovered from cardiac surgery. The developed algorithm showed an overall area under the ROC curve equal to 0.958 for discrimination between different hemodynamic states, thus demonstrating a lot of promise in wearable cardiovascular monitoring. Still, due to a limited and clinically heterogeneous sample, the results cannot be generalised.

Behavioral variables assessed with wearables are also very valuable. For instance, Niemelä et al. (2024) investigated 1,134 participants of the Northern Finland Birth Cohort 1986 with the help of wearable ring measurements of physical activity, sedentary time and sleep. As a result, an

association between 24-hour movement behavior and cardiometabolic health was discovered. In addition, the study by de Vries et al. (2023) focused on heart-rate variability during sleep, thus showing how wearable-based cardiac information can be used for the assessment of physical functioning. These studies show that cardiovascular risk assessment should not rely only on one kind of wearable data, as physiological and behavioural indicators complement each other.

Device development has also produced promising results in recent years. So, Dheman et al. (2024) assessed a wearable chest patch in ten healthy participants to perform heart rate, blood pressure and respiratory-rate monitoring, which showed very good measurement performance; however, a very small healthy sample is a major limitation of external validity. Giordano et al. (2024) also evaluated wrist-based ultrasound heart rate monitoring in ten healthy volunteers and demonstrated a very high agreement with ECG-derived heart rate ($r = .99$), but once again, a very limited sample restricts population-level inference.

Overall, the recent empirical research proves the technical and analytical value of wearable-derived heart rate, blood pressure, sleep and physical activity data. However, there is a lot of methodological heterogeneity related to devices, samples, variables and analysis used. While some studies focus on device validation or prediction, others explore behavioural relationships. Therefore, evidence is fragmented as there is no unified statistical framework to interpret multiple physiological and behavioural indicators and their relationship with cardiovascular risk. This clearly demonstrates the need for the comparative analysis of wearable cardiovascular indicators in the present study.

2.6 Comparison of Empirical Studies

Table 1: Comparison of Studies on Wearable Technologies and Cardiovascular Health

Study	Main Focus	Key Contribution	Limitation
Li et al. (2025)	Digital cardiovascular monitoring	Highlights continuous monitoring and personalised cardiovascular care	Limited empirical statistical assessment
Moshawrab et al. (2023)	Cardiovascular disease detection	Identifies the value of wearable sensors for early detection	Systematic review rather than primary empirical research
Canali et al. (2022)	Wearable data quality	Highlights reliability, interoperability, fairness and health-equity issues	Does not directly analyse cardiovascular risk
Huang et al. (2022)	AI and wearable sensor data	Demonstrates predictive potential of AI using physiological measurements	Limited interpretability of complex models
Kiran et al. (2025)	Explainable cardiovascular risk prediction	Emphasises transparency and interpretability in cardiovascular prediction	Primarily focused on AI framework development
Qadrie et al. (2025)	Wearables in healthcare	Highlights continuous and remote physiological monitoring	Predominantly conceptual rather than empirical
Sultana et al. (2025)	Cardiovascular monitoring and	Demonstrates preventive potential of wearable	Systematic review without direct dataset

	prevention	technologies	validation
Bhaltadak et al. (2024)	Wearables in cardiovascular medicine	Establishes clinical importance of wearable monitoring	Focuses mainly on technological developments

Overall, the current literature shows that wearables could improve continuous cardiac monitoring, early detection, and personalised care (Li et al., 2025). Nevertheless, there is still a significant methodological gap. The AI-related works pay more attention to predictability but have some issues with explainability, while the explainable approaches try to address it (Huang et al., 2022; Kiran et al., 2025). At the same time, some other works focus on the issues of data quality and integration but do not provide any quantitative evidence on the relationships between the indicators and the risk of cardiovascular diseases (Canali et al., 2022). Additionally, some studies are based on review evidence, not on quantitative analysis (Moshawrab et al., 2023; Sultana et al., 2025).

2.7 Research Gap and Study Justification

Existing literature highlights the capacity of wearable technology to continuously monitor cardiovascular health. Nevertheless, there are notable empirical lacunae in the existing body of research. Existing studies tend to focus more on applications of the technology itself, reliability of devices, or even predictions based on artificial intelligence without considering a straightforward statistical analysis of correlations between multiple indicators obtained from wearable devices and cardiovascular risks (Li et al., 2025). Furthermore, review-based literature has already pointed out the clinical significance of continuous monitoring but has not validated these relationships through empirical analysis of wearable data (Moshawrab et al., 2023).

2.8 Chapter Summary

In summary, the literature shows that wearable devices enable constant monitoring of cardiovascular health based on physiological and behavioural markers, but issues surrounding quality of data and interpretation persist. Such limitations have laid the groundwork for the need to conduct transparent statistical analysis of such cardiovascular health risk indicators, as described in Chapter 3.

Chapter 3: Research Methodology

3.1 Introduction

In this chapter, the research methodology used for exploring the relationships between the physiologic and behavioural variables from wearables and cardiovascular disease risk will be introduced. Based on the quantitative nature of the study, this chapter elaborates on the research philosophy, deductive research approach, observational study design, database selection, sampling of participants, and variable selection, as well as the data preparation, Python-based analysis, hypothesis testing, reliability and validity, ethics, and methodological constraints.

3.2 Research Philosophy

Positivist research philosophy is used in this research since the cardiac parameters collected through wearables are objectively measured and analysed statistically. The positivist research philosophy is concerned with the analysis of objective numerical data with minimal bias on the part of the researcher. It is for this reason that the positivist research philosophy is applicable in analysing the measurable variables, including the statistical relationship between the variables.

3.3 Research Approach

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3.4 Research Design

In this study, an approach involving quantitative, secondary observational research design was employed to analyse the relationship between physiological and behavioural measures and cardiovascular risks. The choice of a quantitative research design in this study was justified by the presence of variables that could be measured numerically. The use of an observational design implied that no manipulation of participants was conducted in this case, whereas the use of a secondary data analysis technique made it possible to analyse existing wearable health records.

3.5 Research Strategy

An explainable statistical approach was utilised in the study for analysing the relationship between physiological parameters and cardiovascular risk. The statistical approach used in the study included data pre-processing, exploration and analysis, description of statistics, test of normality, correlation, group comparison tests and hypothesis testing using Python. The explainable statistical approach was preferred in favour of black-box prediction as it emphasised interpretability of statistical results.

3.6 Dataset Source, Description and Selection

The research adopted a secondary wearables dataset that was chosen to explore possible relationships between physiological, behavioural, and demographic metrics and cardiovascular risks. The dataset was made up of 5,000 measurement-level entries belonging to 1,000 different individuals, who had five measurements each with 18 variables. The data were measured for the period ranging from January through December 2025. The variables used included age, gender, body mass index, resting and mean heart rates, number of steps taken, number of minutes of

physical activity, hours slept, SpO₂, energy expended, systolic and diastolic blood pressures, cardiovascular risk score, and cardiovascular risk level. The selection was appropriate since these variables offer complementary metrics about cardiovascular health and behaviour, which are derived from wearables. The original methodology mentioned PhysioNet as the database repository. The specific dataset name, DOI, country, and wearable device brand should be specified only if they are confirmed in the dataset documentation.

3.7 Population, Sample and Variables

The target population was defined as those whose physiological and behavioural health measures were captured via wearable device readings. The measurements taken via wearables can be used as objective measures that may be pertinent to the cardiovascular analysis, such as heart rate, physical activity and related physiological measures (Alimbayeva et al., 2024). In the original dataset, there were 5,000 records of measurements from 1,000 unique subjects with five measurements each. Aggregating the measurements at the participant level to avoid assuming independence for the repeated measures ensured the use of a total of 1,000 participants in the analysis. The selection of variables was based on the physiological pertinence of the variables to cardiovascular risk analysis and study objectives.

Table 2: Variables Used in the Statistical Analysis

Variable	Role	Measurement
Age	Predictor/control	Years
Gender	Control	Female/Male
BMI	Predictor	kg/m ²
Resting Heart Rate	Predictor	bpm

Average Heart Rate	Predictor	bpm
Average Daily Steps	Predictor	Steps/day
Active Minutes	Predictor	Minutes/day
Sleep Duration	Predictor	Hours
SpO ₂	Predictor	%
Calories Burned	Predictor	kcal
Resting Energy Expenditure	Predictor	kcal
Active Energy Expenditure	Predictor	kcal
Systolic Blood Pressure	Predictor	mmHg
Diastolic Blood Pressure	Predictor	mmHg
Cardiovascular Risk Score	Dependent	Continuous risk score
Risk Category	Grouping/outcome	Low/Moderate/High

3.8 Data Collection and Integrity Verification

The secondary data was obtained from the selected repository and then imported into Python using Pandas for analysis. After importing, data validity was checked based on the number of variables and their attributes such as the name, types, units of measurement, and completeness of records. This process guaranteed that the data set was imported without errors.

3.9 Data Pre-processing and Cleaning

There was a total of 5,000 records for measurements on 1,000 participants, with 5 measurement data available per participant in the initial dataset. Since inference is performed at participant-level, multiple measurement data were aggregated into single participant observations. Aggregation of continuous behavioural and physiological variables was done with aggregation

methods programmed in the Python script, while the participant-level variables were kept consistent throughout. This reduced the size of the dataset to be used for analysis from 5,000 records of measurements to 1,000 participant-level observations. This was because aggregation of the data avoided treating multiple observations from the same participant as separate participants.

3.10 Python-Based Analytical Framework

Analysis was carried out using Python via Google Colab, giving a reproducible platform for data and statistical analysis of the wearable device dataset. The process of analysis built on the already established structure in the methodology, including data preparation, exploratory analysis, statistical analysis, and visualisation. Pandas and NumPy were used in the analysis to manipulate and conduct calculations, while SciPy was used for conducting statistical tests. On the other hand, Matplotlib was used to create visualisations. The analysis involved importing the dataset, checking its integrity, preparing the data, aggregating at the level of each individual participant, conducting descriptive statistics, testing normality, correlation, comparing groups, and hypothesis testing.

3.11 Statistical Analysis and Research Question Alignment

The data analysis was done using Python to explore correlations between physiologic and behavioural measures captured from wearables and cardiovascular risk. The first step was to calculate descriptive statistics in the form of mean, median, standard deviation, minimum, and maximum. This is an extension of the statistical method that was presented in the previous section of the methodology. The Shapiro–Wilk test was applied to assess the normality of continuous data. Pearson’s correlation was employed to measure linear correlation if the underlying assumptions held true. For variables that did not follow a normal distribution, Spearman’s rank correlation coefficient was used. Differences in the cardiovascular-risk categories such as Low, Moderate, and

High were analysed using ANOVA and Kruskal–Wallis tests. The chi-square test was conducted to investigate the relationship between gender and risk category. All the hypotheses were tested at a 95% confidence level ($\alpha = 0.05$); thus, $p < 0.05$ was statistically significant.

The appropriate statistical methods used in the study were chosen based on measurement features and distribution of the variables at a participant level. For normality testing, the Shapiro-Wilk test at a significance level of $\alpha = .05$ was employed. Several principal variables did not show a normal distribution; that is why Spearman's rho was the preferred way to assess monotone relationship between physiological-behavioural measures and cardiovascular risk score. This method does not assume normal distribution of variables and can determine both strength and direction of association. To compare indicators that were continuous variables between independent groups of cardiovascular low, moderate, and high risk categories, the Kruskal-Wallis H test was used due to lack of support for parametric assumptions. For comparison of two categorical variables, gender and risk category, the chi-square test of independence was used.

3.12 Reliability and Validity

Reliability was enhanced by employing an established and constant analytical process in Python involving identical processing steps, statistical methods and an alpha of 0.05 on the entire dataset. This reproducible process helps minimise researcher bias and makes it possible to replicate the process. Internal validity was confirmed by conducting quality controls on the data, using clinical judgment in identifying outliers and choosing proper statistical techniques based on the normality of the data. External validity is low since the results are based on secondary data that may not be applicable to other people, healthcare facilities and wearables.

3.13 Ethical Considerations

The data that was used in this research study was secondary data, and it was taken from the publicly available database called PhysioNet; hence, there was no participant involvement in any form. It was ensured that the participants' privacy would remain intact because no personal details of the participants were used in the analysis. The data was analysed only for academic purposes while keeping in mind the principles of GDPR concerning privacy.

3.14 Methodological Limitations

Limitations of the methodology are associated with the use of the secondary source of data, implying that there is no possibility to control for the sample selection and measurement methods. Moreover, wearables can also be associated with possible measurement errors caused by the sensors, influencing physiological measurements (Alimbayeva et al., 2024). In addition, the chosen approach allows finding correlations between variables only.

3.15 Chapter Summary

In this chapter, the framework was developed to analyse the relationships between the physiological markers collected through wearables and the risk of heart disease in a quantifiable and deductive manner. The study used secondary data along with proper preprocessing of the data with statistical tests using Python. Issues related to reliability, validity, and ethics were considered as well.

Chapter 4: Results and Analysis

4.1 Introduction

This chapter discusses the findings obtained through empirical research related to the cardiovascular wearable data analysis. The initial data set consisted of 5,000 records at the level of individual measurements for 1,000 distinct individuals, and thus it included 1,000 records at the level of participant. The chapter provides data about quality and description, cardiovascular risk distribution, normality test, correlation analysis, differences between the groups of risk, and hypothesis testing. The level of statistical significance was $\alpha = 0.05$.

4.2 Data Quality and Pre-processing Results

The initial dataset had 5,000 measurement-level observations on 18 variables, with a total of 1,000 individuals, where there were precisely five measurement observations for each participant. Preliminary quality analysis showed no missing observations and no duplicate rows. Yet, there were 33 duplicate participant-date observations. Physiological plausibility checking indicated that the variables had plausible values in their predefined physiological ranges except for 20 SpO₂ observations (0.4%), which fell outside the predefined 70–100%. Interquartile range analysis revealed statistical outliers on several variables, such as 73 diastolic blood pressure observations (1.46%) and 39 systolic blood pressure observations (0.78%) (Uddin, 2024). The statistically plausible extremes were kept, as we do not want to lose any information on cardiovascular risks in this way. Finally, aggregation of the repeated measures led to a participant-level dataset consisting of 1,000 observations. There was no need for normalisation of data for the chosen statistical analyses.

4.3 Participant Characteristics and Descriptive Statistics

After aggregating the repeated measurements, the dataset at the level of the individual consisted of 1,000 participants. The mean age was 49.78 years (SD = 17.42), and the mean BMI was 26.26 kg/m² (SD = 4.14). The mean values for the cardiovascular parameters were 73.10 bpm for the resting heart rate and 88.83 bpm for the average heart rate. Mean systolic and diastolic blood pressures were 125.29 mmHg and 79.57 mmHg, respectively.

The average number of steps per day and minutes of being active per day were 10,580 and 62.79, respectively, showing that individuals were physically active. The mean sleep duration was 7.03 hours, while the mean SpO₂ was 97.02%. The average values for resting and active energy expenditure were 1,811.56 kcal and 434.35 kcal, respectively. The average value of total energy expenditure was 4,717.31 kcal. The mean cardiovascular risk score was 42.38 (SD = 13.36). In general, it is possible to see that there is a large variation in age.

4.4 Cardiovascular Risk Distribution

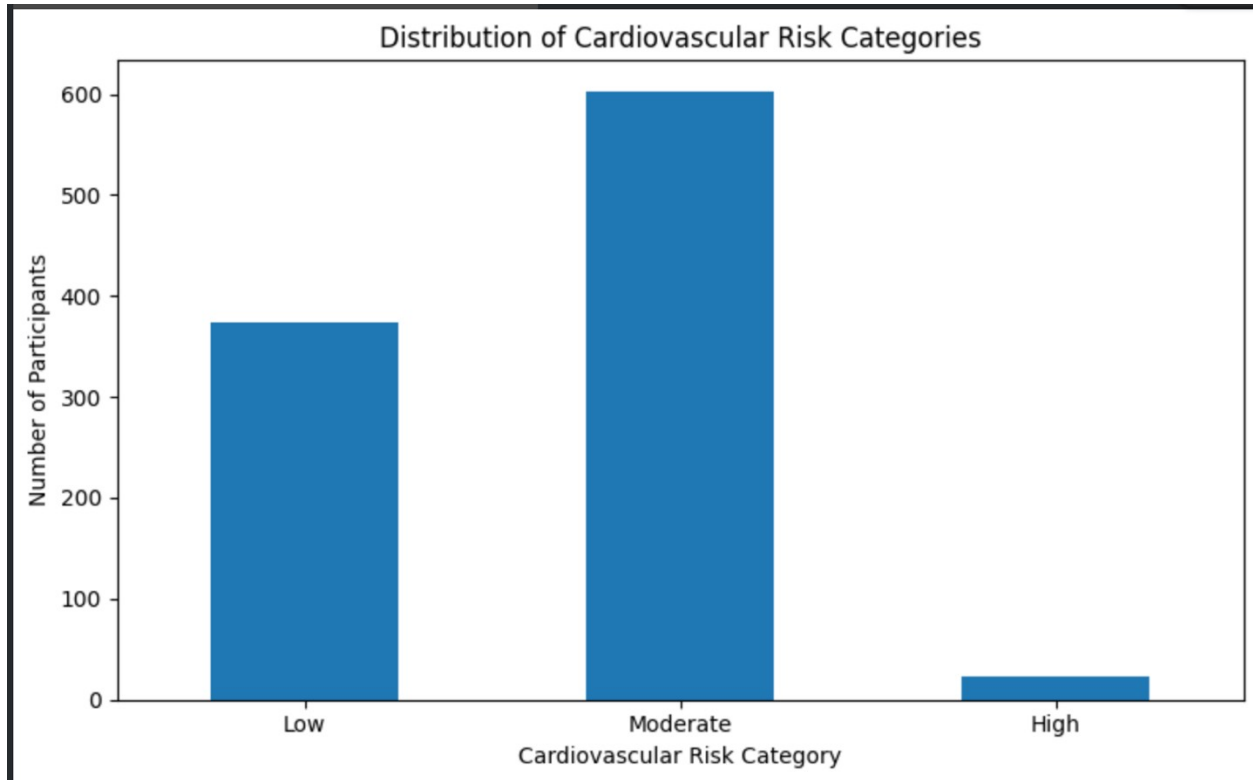


Figure 1: Distribution of Cardiovascular Risk Categories

(Source: Self-created)

Based on the participant level, it was found that the distribution of cardiovascular risk across the three categories was not equal. In a total of 1,000 participants, 374 (37.4%), 603 (60.3%) and 23 (2.3%) were identified to have low, moderate and high risk, respectively. This meant that there was a predominance of moderate risk of cardiovascular problems in the sample because it comprised more than half of all the individuals who were analysed. The high-risk cases, however, made up a small percentage of the participants. Figure 4.1 below demonstrates the pattern of distribution in this case. The mean risk scores for the participants also increased steadily with an increase in category: 28.532 for low risk, 49.836 for moderate risk and 72.097 for high risk.

4.5 Normality Assessment

The Shapiro-Wilk test was conducted at the 0.05 significance level to examine the distribution of the continuous variables. Active minutes per day ($p = .698$), sleep duration ($p = .113$), SpO_2 ($p = .538$), resting energy expenditure ($p = .925$) and active energy expenditure ($p = .959$) demonstrated an approximate normal distribution. In addition, age, body mass index, resting heart rate, average heart rate, number of daily steps, calories burnt, systolic blood pressure, diastolic blood pressure, and cardiovascular risk score showed significant results in Shapiro-Wilk test ($p < .05$). Since most important cardiovascular variables, including the dependent variable, did not meet the assumption of normal distribution, the priority was given to non-parametric analysis (Bharath & Sheeba (2026)). The Spearman's rank correlation coefficient was chosen to examine the association between variables and cardiovascular risk score, while the Kruskal-Wallis test was selected for group comparison.

4.6 Association Between Wearable Measures and Cardiovascular Risk

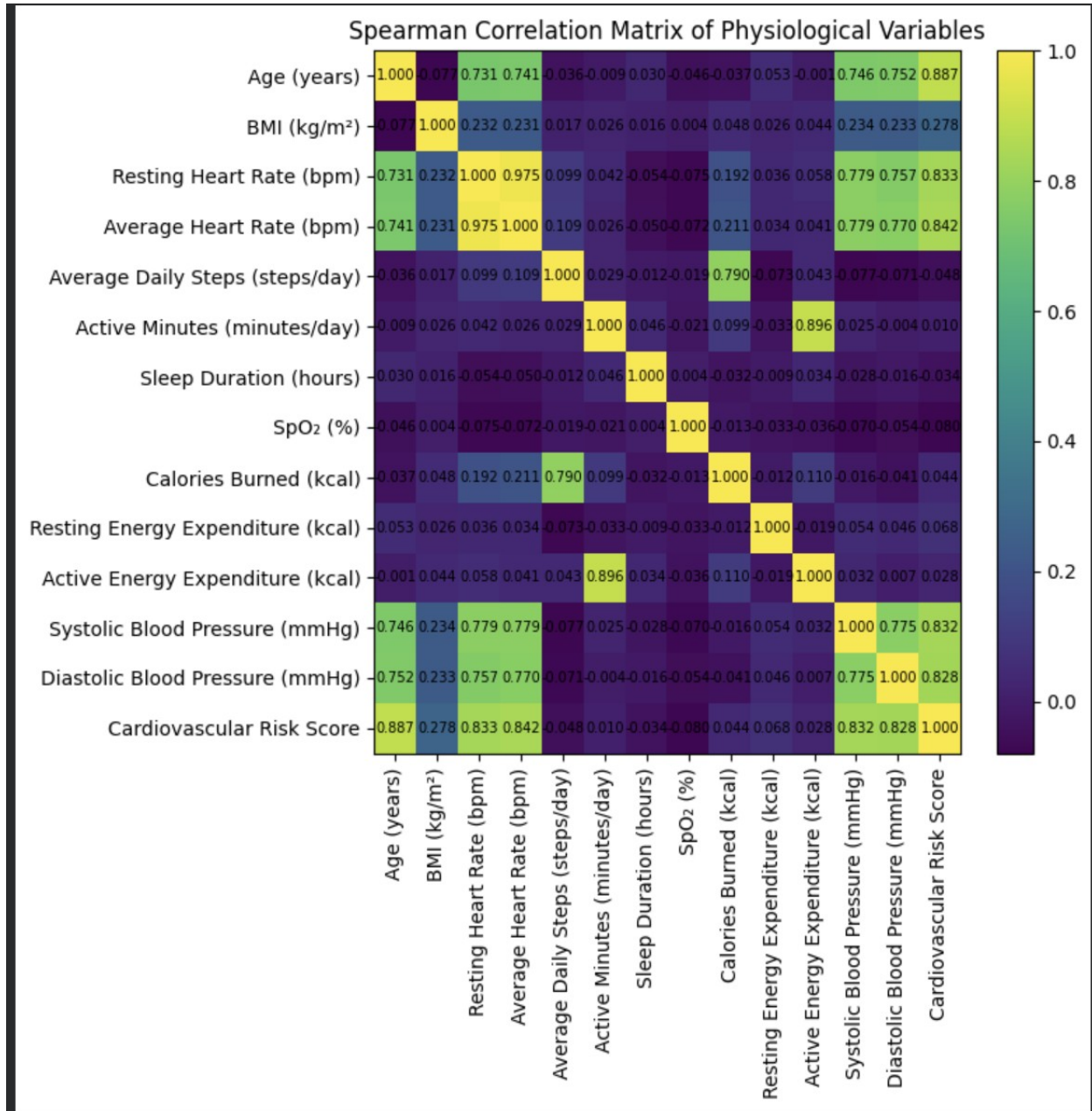


Figure 2: Spearman Correlation Heatmap

(Source: Self-created)

Spearman's rank correlation analysis was performed to assess any associations between the derived variables from wearables and cardiovascular risk score. It was found that age showed the highest positive correlation with cardiovascular risk ($\rho = .887$, $p < .001$), which means that

increased age was highly correlated with increased risk score. Significant strong positive correlations were also found for the average heart rate ($\rho = .842, p < .001$), resting heart rate ($\rho = .833, p < .001$), systolic blood pressure ($\rho = .832, p < .001$), and diastolic blood pressure ($\rho = .828, p < .001$). This result suggests that age, heart rate measures, and blood pressure measures are the variables showing the strongest association with cardiovascular risk among the participants' data.

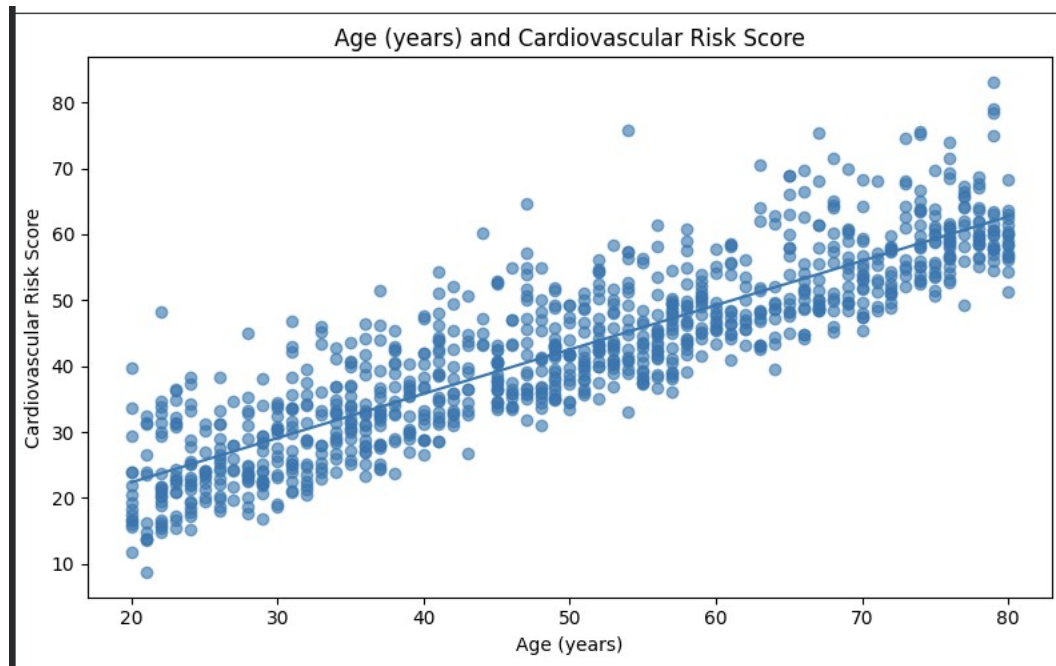


Figure 3: Age vs Cardiovascular Risk Score scatterplot

(Source: Self-created)

BMI has demonstrated a significant, although not very strong, positive correlation with cardiovascular risk ($\rho = .278, p < .001$). At the same time, SpO_2 showed a significant weak negative correlation with cardiovascular risk ($\rho = -.080, p = .011$), while the resting energy expenditure showed a weak positive correlation ($\rho = .068, p = .032$). Despite their statistical significance, the low values of the correlation coefficients suggest that their associations were not as strong as those for age, heart rate, and blood pressure.

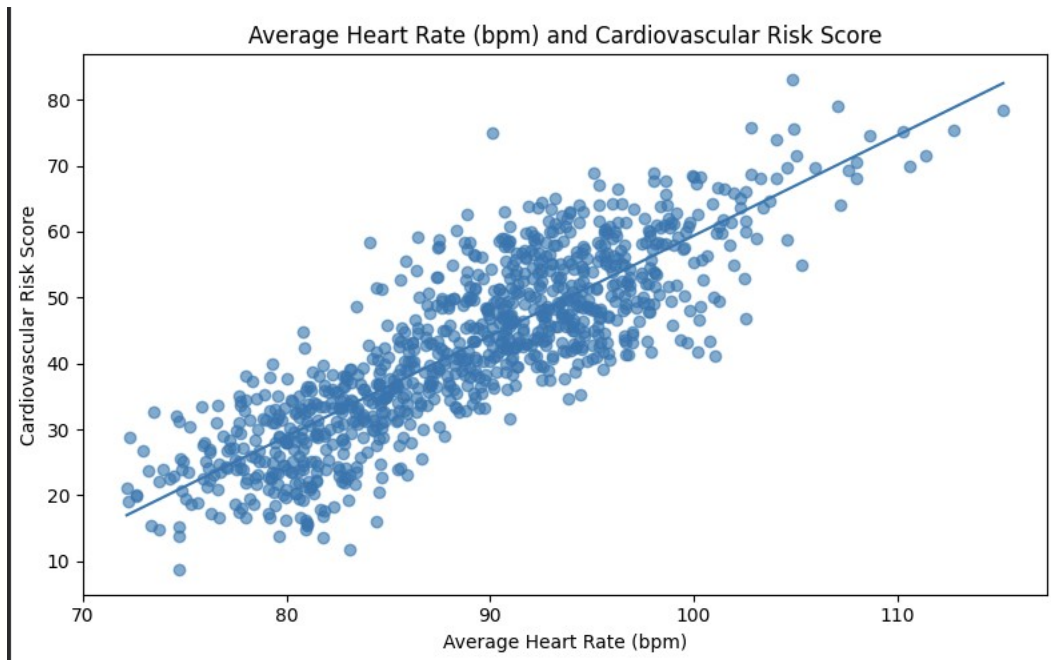


Figure 4: Average Heart Rate vs Cardiovascular Risk Score scatterplot

(Source: Self-created)

No significant correlation was found for several wearable-derived behavioural and energy measures in relation to cardiovascular risk. These measures include average daily steps ($\rho = -.048$, $p = .128$), calories burned ($\rho = .044$, $p = .163$), sleep duration ($\rho = -.034$, $p = .279$), active energy expenditure ($\rho = .028$, $p = .371$), and active minutes per day ($\rho = .010$, $p = .753$). Figure 4.2 depicts the correlation heatmap for Spearman's test. As can be seen from it, the correlations are concentrated mostly around age, heart-rate measures, and blood pressure.

4.7 Cardiovascular Measures Across Risk Categories

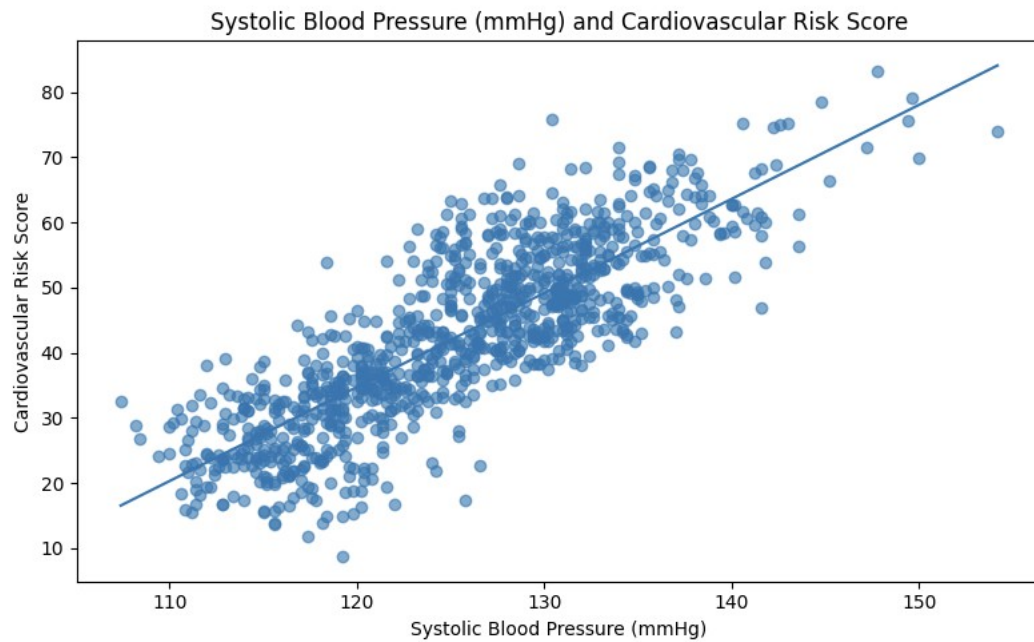


Figure 5: Systolic BP by Risk Category boxplot

(Source: Self-created)

The comparison among low-, moderate-, and high-risk groups revealed some considerable distinctions in terms of a number of cardiovascular parameters. The mean age of participants dramatically increased from 32.64 years in the low-risk group to 59.59 years in the moderate-risk group and 71.22 years in the high-risk group. In the same manner, the mean values of resting heart rate (67.16, 76.29 and 86.29 bpm) and average heart rate (81.45, 92.81 and 104.67 bpm) progressively increased in each subsequent risk category. The increase in the mean blood pressure values is also obvious: the mean systolic blood pressure values increased from 117.65 mmHg to 129.41 mmHg and 141.60 mmHg; the diastolic pressure values increased from 74.78 mmHg to 82.12 mmHg and 90.35 mmHg, correspondingly. Finally, the mean value of cardiovascular risk score increased from 28.53 to 49.84 and 72.10.

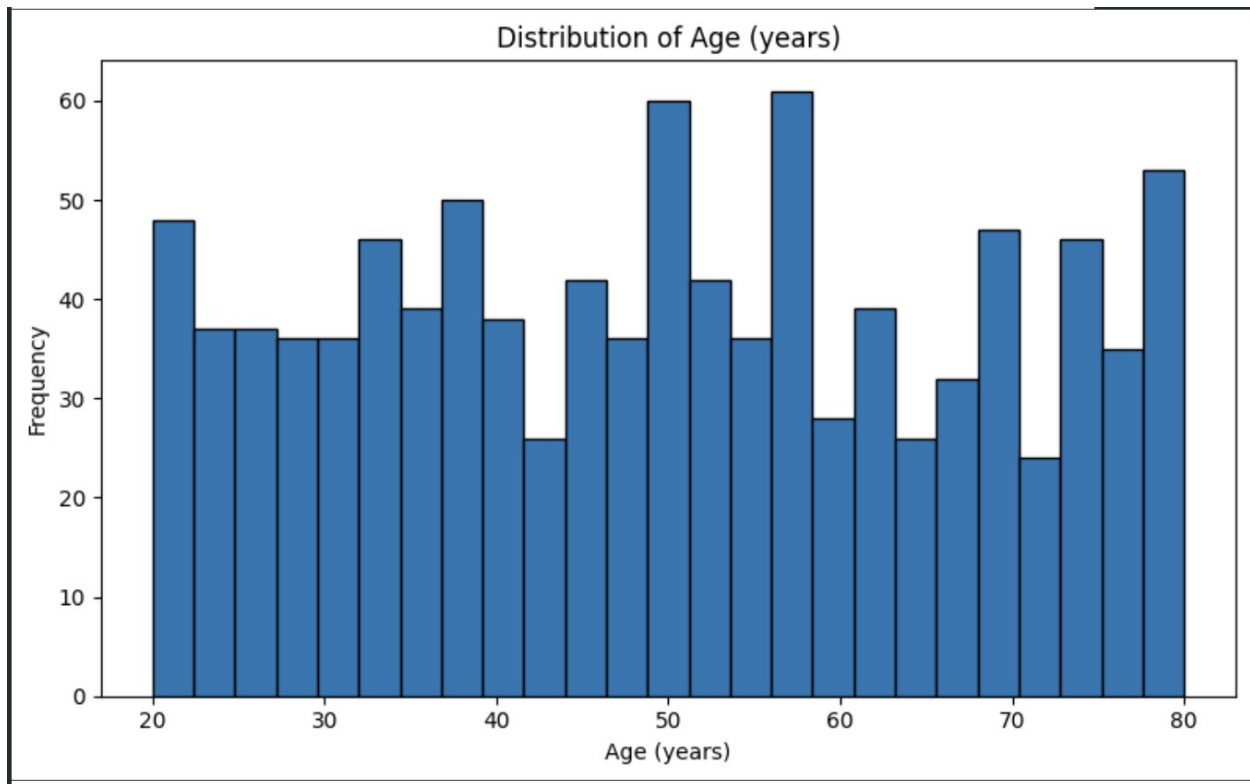


Figure 6: Age by Risk Category boxplot

(Source: Self-created)

Kruskal–Wallis testing revealed statistically significant differences between the risk categories in terms of age ($H = 597.590$, $p < .001$), BMI ($H = 76.790$, $p < .001$), resting heart rate ($H = 636.700$, $p < .001$), average heart rate ($H = 647.970$, $p < .001$), systolic blood pressure ($H = 654.724$, $p < .001$) and diastolic blood pressure ($H = 663.161$, $p < .001$). Steps per day varied significantly as well ($H = 6.250$, $p = .044$) but the significance of the difference was much lower. Active minutes per day, hours slept per night, oxygen saturation, number of burned calories, resting and active energy expenditure did not differ statistically significantly

4.8 Gender and Cardiovascular Risk

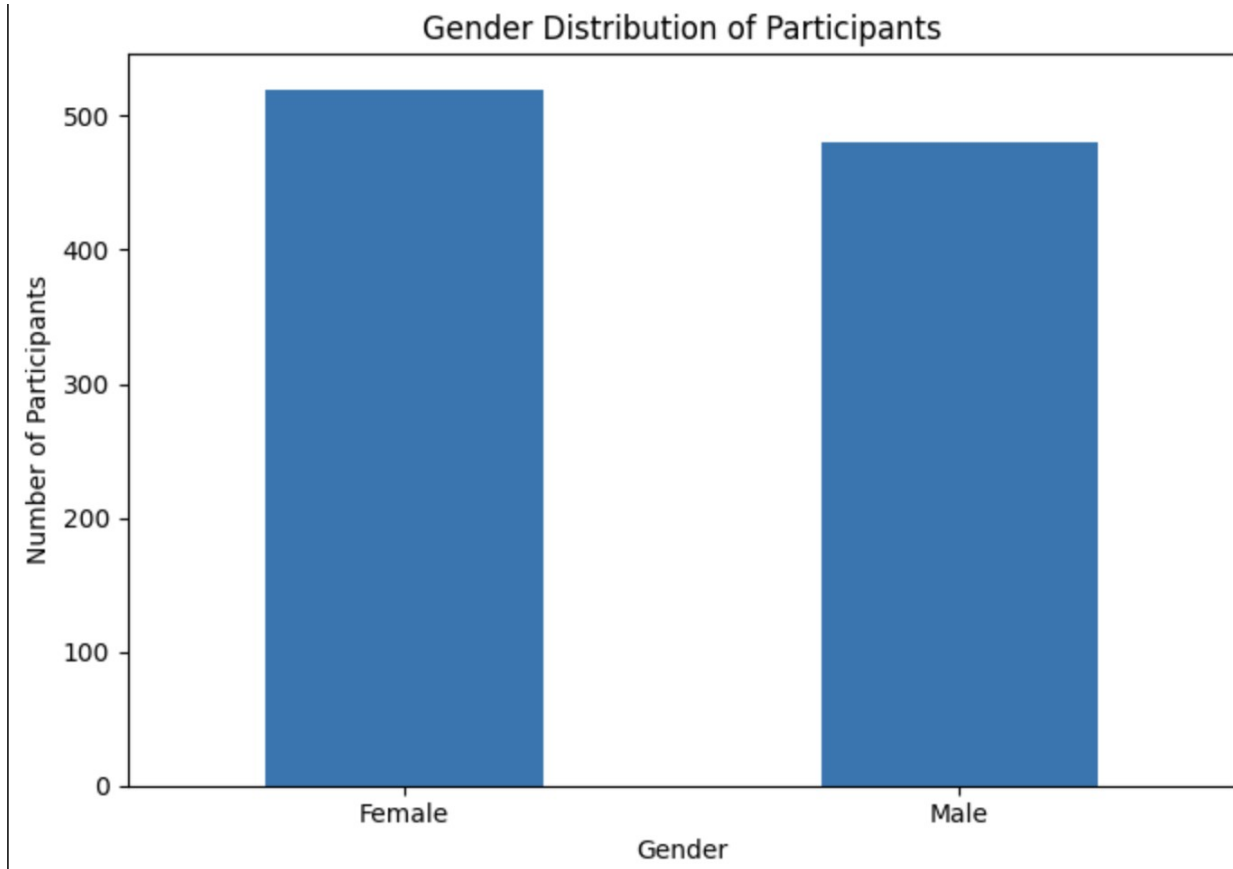


Figure 7: Gender Distribution

(Source: Self-created)

The gender ratio was evenly distributed among the 1,000 subjects, where 520 were female (52.0%), and 480 were male (48.0%). Among the cardiovascular risk categories, there were 201 low-risk females, 305 moderate-risk females, and 14 high-risk females as opposed to 173 low-risk males, 298 moderate-risk males, and 9 high-risk males. A test of independence using the chi-square statistic showed that there was no statistically significant relationship between gender and cardiovascular risk categories, $\chi^2(2) = 1.667$, $p = .434$. As such, there was no rejection of the null hypothesis, given that the value was above the threshold of 0.05.

4.9 Hypothesis Testing and Research Question Findings

Hypothesis testing was performed at the 5% significance level ($\alpha = 0.05$) using Spearman's rank correlation, and the null hypothesis was rejected when $p < 0.05$. Table 4.8 below presents the findings.

Table 3: Hypothesis Testing Summary

Predictor	Spearman's ρ	p -value	Decision
Age	0.887	<0.001	Reject H_0
Average Heart Rate	0.842	<0.001	Reject H_0
Resting Heart Rate	0.833	<0.001	Reject H_0
Systolic BP	0.832	<0.001	Reject H_0
Diastolic BP	0.828	<0.001	Reject H_0
BMI	0.278	<0.001	Reject H_0
SpO ₂	-0.080	0.011	Reject H_0
Resting Energy Expenditure	0.068	0.032	Reject H_0
Average Daily Steps	-0.048	0.128	Fail to reject H_0
Calories Burned	0.044	0.163	Fail to reject H_0
Sleep Duration	-0.034	0.279	Fail to reject H_0
Active Energy Expenditure	0.028	0.371	Fail to reject H_0
Active Minutes per Day	0.010	0.753	Fail to reject H_0

The study has answered the research question as it shows that the most significant associations were between cardiovascular risk and demographic variables as age and cardiovascular measures,

including heart rate and blood pressure. The highest association was seen for the age variable ($\rho = 0.887$), then the average heart rate ($\rho = 0.842$), the resting heart rate ($\rho = 0.833$), the systolic blood pressure ($\rho = 0.832$), and the diastolic blood pressure ($\rho = 0.828$). BMI showed a significant positive association, while SpO_2 showed a negative, though weak, association (Sarad et al., 2026, April). However, daily steps, burned calories, the duration of sleep, and activity measures have not shown any statistically significant associations.

4.10 Chapter Summary

The chapter has provided analysis on the risk factors of cardiovascular disease in relation to 1,000 subjects. As per the findings of the study, age ($\rho = 0.887$), mean heart rate ($\rho = 0.842$), resting heart rate ($\rho = 0.833$), systolic blood pressure ($\rho = 0.832$) and diastolic blood pressure ($\rho = 0.828$) have been found to be the most significant factors relating to the risk factor score. Among the three levels of risk factor, moderate risk (60.3%) is the highest level of risk followed by low risk (37.4%) and high risk (2.3%). Gender has not shown any significant relationship with risk level ($p = 0.434$).

Chapter 5: Conclusion and Recommendations

5.1 Introduction

This chapter ends the research by summarising the key results on wearable health indicators and cardiovascular risk. It will examine how well the research aims have been met, the applicability and limitations of the research, as well as recommend further studies before making the conclusions.

5.2 Summary of Key Findings

A total of 1,000 participant-level data sets, out of 5,000 wearable data sets, were used for the analysis. The cardiovascular risk assessment indicated that 60.3% of the participants had a moderate-risk classification, whereas 37.4% of the participants had low risk, and only 2.3% had high risk. Correlation analysis using Spearman's correlation indicated age ($\rho = .887$) to be the most correlated factor with the cardiovascular risk score, followed by the average heart rate ($\rho = .842$), resting heart rate ($\rho = .833$), systolic blood pressure ($\rho = .832$), and diastolic blood pressure ($\rho = .828$). BMI, SpO₂, and resting energy expenditure also showed significant correlations. On the other hand, daily steps, calories burnt, sleep time, active energy expenditure, and active minutes did not show any significant correlation with the cardiovascular risk score. There were also significant differences between risk groups based on major physiological indicators. Lastly, there was no significant relationship between gender and cardiovascular risk, $\chi^2 (2) = 1.667$, $p = .434$.

5.3 Achievement of Research Aim and Objectives

The experiment was successful in achieving its purpose of finding correlations between wearables' health indicators and cardiovascular risk. Descriptive analysis helped to define the physiological,

demographic, and lifestyle features of 1,000 people. Correlation analysis found out that age ($\rho=0.887$), average heart rate ($\rho=0.842$), resting heart rate ($\rho=0.833$), systolic blood pressure ($\rho=0.832$), and diastolic blood pressure ($\rho=0.828$) were the main factors correlated with cardiovascular risk. Risk group comparison helped to identify significant differences in some indicators between people who belonged to different risk groups (Babalola et al., 2025, April). At last, hypothesis testing helped to find out the statistically significant relationships and non-statistical relationships.

5.4 Practical and Research Implications

The results indicate the possible use of wearable physiological data in detecting correlations with cardiovascular risks. Age proved to have the most significant correlation with cardiovascular risks, while the average heart rate, resting heart rate, systolic blood pressure, and diastolic blood pressure also had strong correlations with the cardiovascular risk scores. The results indicate that physiological parameters could potentially serve for cardiovascular health tracking and cardiovascular risk-oriented research. But it is important to remember that the correlations detected are not cause-and-effect relationships, and they cannot diagnose a medical condition.

5.5 Limitations

Various limitations must be recognised. First, the research is based on secondary data through observation; thus, the discovered correlations can neither prove nor disprove causation between the wearable factors and the risk of heart diseases. Even though there was an availability of 5,000 observations, their aggregation into one observation per each person results in losing some information regarding variations in time for each participant. Also, it must be mentioned that the

risk groups have been highly imbalanced; for instance, only 23 participants (2.3%) were high risk compared to 603 (60.3%) in the moderate-risk category.

5.6 Recommendations and Future Research

In future studies, it is necessary to study larger and more heterogeneous populations so that the results will be more generalizable, especially through the inclusion of a larger number of high-risk individuals. Longitudinal studies need to be conducted in order to assess the changes in time in the wearable biomarkers and in the cardiovascular risk levels. The external validation of the results through an independent data set is also needed.

5.7 Final Conclusion

Overall, it can be noted that the highest connection is seen in relation to age, heart rate and blood pressure parameters of the wearable data set. In this regard, the statistical identification of physiological features, which have the highest connection with cardiovascular risk, helps to prove that wearable data has potential in cardiovascular risk assessment.

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